

1) Create a table listing the main functions and typical ports used by switches, routers, hubs, firewalls, and access points in a home WiFi setup.

Network Devices Comparison

- Switch — Function: Connects multiple devices within a LAN and forwards frames between ports. Typical ports: Ethernet ports and console/management port.
- Router — Function: Connects different networks and routes IP packets; commonly provides the gateway to the Internet. Typical ports: WAN/Internet, LAN Ethernet, and console/management port.
- Hub — Function: Connects devices in a basic LAN and repeats incoming data to its other ports. Typical ports: Ethernet ports and power.
- Firewall — Function: Filters and controls network traffic according to security rules. Typical ports/interfaces: WAN/Internet, LAN interfaces, and management interface.
- Access Point — Function: Provides wireless network access and connects WiFi clients to the wired LAN. Typical ports: Ethernet, power, and antennas (depending on model).

Summary Table

- Switch: LAN connectivity — Ethernet, Console.
- Router: Inter-network routing/gateway — WAN, LAN, Console.
- Hub: Basic signal repeating — Ethernet, Power.
- Firewall: Traffic filtering/security — WAN, LAN, Management.
- Access Point: Wireless connectivity — Ethernet, Power, Antenna.

2) Given images or diagrams of network devices, label each device and mark the ports (Ethernet, console, power, antenna) for at least three devices.

Step 1: Identify the Devices

- Identify the device from its shape, labels, and visible interfaces.
- Use a switch, router, and wireless access point as the three example devices.

Step 2: Label the Switch

- Ethernet ports — used to connect PCs, printers, routers, and other LAN devices.
- Console port — used for local device configuration and management.
- Power — supplies electrical power to the switch.

Switch Diagram



Step 3: Label the Router

- WAN/Internet port — connects the router toward the Internet or an upstream network.
- LAN Ethernet ports — connect local wired devices.
- Console port — used for configuration and management on supported routers.
- Power — supplies electrical power.

Router Diagram



Step 4: Label the Access Point

- Ethernet port — connects the access point to the wired network.
- Power — supplies electrical power, either directly or through PoE on supported devices.
- Antenna — transmits and receives wireless signals; some access points use internal antennas.

Access Point Diagram



3) Pick a popular app you use daily (like Instagram, Zomato, or Paytm) and describe in 3–4 lines how at least two different network devices play a role when you use that app.

Example App: Instagram

- When you open Instagram, your phone connects to a WiFi access point or mobile network to reach the Internet.
- The access point provides the wireless connection and forwards traffic to the local network.
- A router routes the traffic between the local network and the Internet so the phone can reach Instagram's servers.
- A firewall may inspect or filter network traffic according to configured security rules.

Network Devices Involved

- Access Point — provides WiFi connectivity to the phone.
- Router — forwards traffic between the local network and the Internet.
- Firewall — can filter and control traffic for security.

4) Suppose your college WiFi is slow and sometimes disconnects. Based on the features of access points, switches, and routers, suggest one possible reason for the issue and how upgrading one device could help.

Possible Reason

- The access point may be overloaded because too many students are connected at the same time, or its wireless coverage and capacity may be insufficient for the area.
- Heavy traffic can also place extra load on the router or switch, but an access point capacity or coverage issue is one likely cause.

Suggested Upgrade

- Upgrade the access point to a newer, higher-capacity model that can support more simultaneous clients and better wireless coverage.
- Place or add access points appropriately so weak-signal areas receive stronger coverage.
- This can reduce congestion and improve connection stability and overall WiFi performance.

Verification

- Check whether disconnects are concentrated in crowded areas or at busy times.
- Compare the number of connected clients and traffic load before and after the upgrade.
- Confirm that users experience better coverage and fewer disconnections.